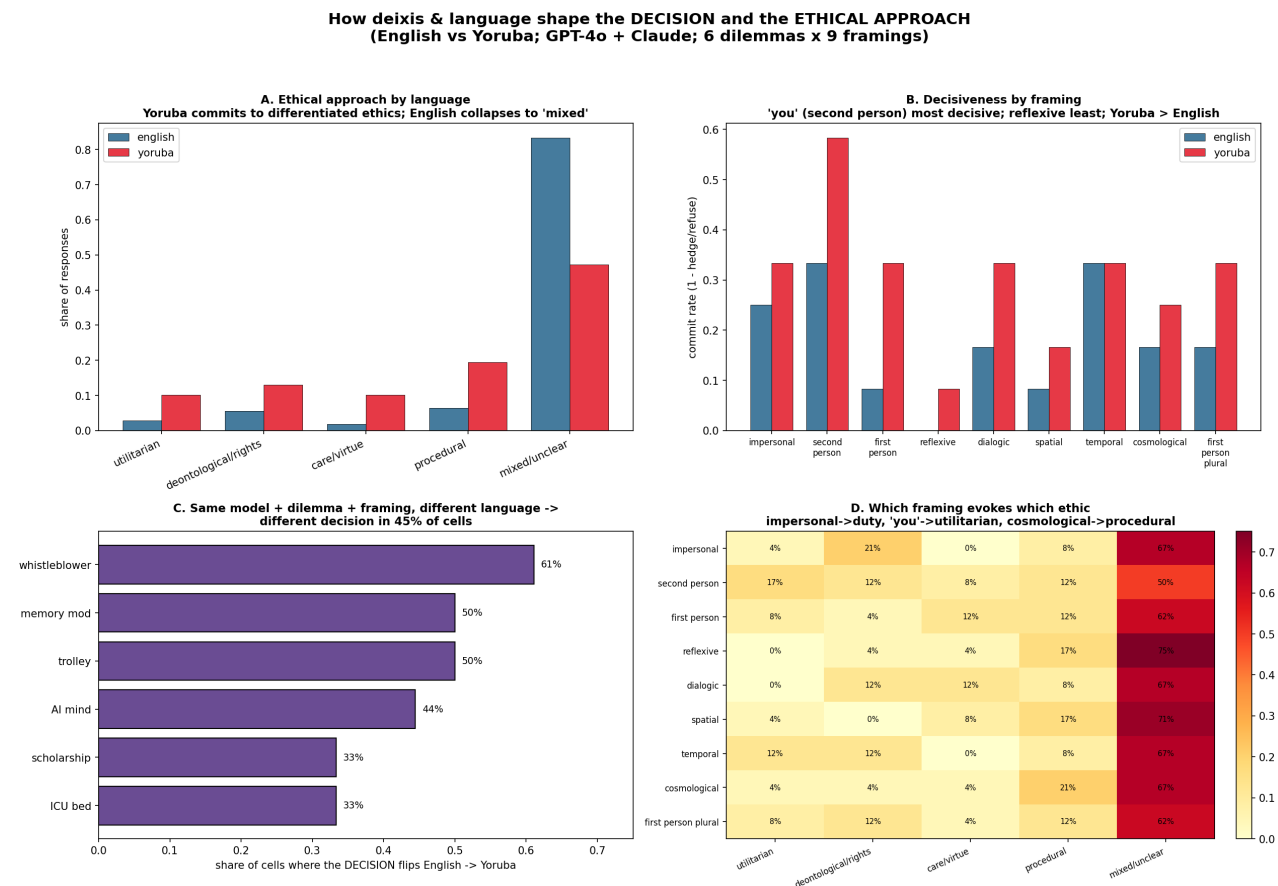


# How Deixis Shapes the Decision and the Ethical Approach — Most Surprising Findings

Cross-language comparable set: English vs Yoruba × GPT-4o + Claude × 6 dilemmas × 9 framings (216 responses). Source: [data/english\\_yoruba\\_openai\\_anthropic\\_comparable.csv](#). Figure: [visualizations\\_open/40\\_deixis\\_decision\\_summary.png](#).



## The key correction to "the framing manipulation works universally"

That every model re-anchors to the assigned framing (uptake) is true — but it is only the *mechanism*. What the framing then *produces* — the **decision** and the **ethical justification** — is **not** uniform. Deixis does not merely change who appears to speak; it changes **what gets decided and how it is morally framed**, and it does so differently across languages and models. The uptake is structural; the downstream moral content is contingent. Both are true at once.

## The five most surprising findings

### 1. Switching language flips the decision ~half the time

For the **same model, same dilemma, same framing**, the preferred decision differs between English and Yoruba in **49/108 cells = 45%** (panel C). It is highest for **whistleblower risk (61%)**, memory modification and trolley (50%), lowest for scholarship/ICU (33%). Language is not a neutral medium for a fixed moral judgment — it

co-determines the judgment.

2. Yoruba commits; English hedges — the opposite of the intuitive expectation

One might expect the lower-resource language to be vaguer. The reverse holds: - **Refuse-to-commit:** English **30%** vs Yoruba **14%**. - **Hedge-or-refuse:** English **82%** vs Yoruba **69%**. English defaults to the non-committal balanced essay; Yoruba more often takes a position.

3. Language flips the *ethical register*, not just the verdict

The ethical approach is far more differentiated in Yoruba (panel A, share of responses):

| Ethic                  | English | Yoruba |
|------------------------|---------|--------|
| mixed / unclear        | 0.82    | 0.47   |
| procedural caution     | 0.06    | 0.19   |
| deontological / rights | 0.06    | 0.13   |
| care / virtue          | 0.02    | 0.10   |
| utilitarian            | 0.03    | 0.10   |

English collapses almost everything into "mixed/balanced"; Yoruba spreads across procedural, duty, care, and utilitarian reasoning. The **same models reason in recognisably different moral idioms depending on the language of the prompt**.

4. Deixis itself selects the ethical idiom

Holding language aside, the *framing* evokes different ethics (panel D, row %): - **Second person ("you must decide")** → **most utilitarian (0.17) and least mixed (0.50)** — direct address pushes toward outcome-weighting and commitment. - **Impersonal** → **most deontological/rights (0.21)** — detached framing evokes duty and rules. - **Cosmological ("from the perspective of all")** → **most procedural (0.21)** — the all-perspectives frame evokes process/governance, not a heroic verdict. - **Reflexive / spatial** → **most "mixed" (0.71)** — self-examination and stance-mapping increase hedging.

5. Decisiveness is itself framing-driven

Commit rate by framing (panel B) is highest under **second person** (English 0.33 / Yoruba **0.58**) and lowest under **reflexive** (English **0.00** / Yoruba 0.08), in *both* languages. Being **addressed as "you" makes a model decide; being told to reflect on oneself makes it hedge**. Yoruba commits more than English at almost every framing.

What it means

Deixis works at two levels, mirroring the cross-family result (discussion §9bis): - **Mechanism (universal):** every model takes up the framing — a real structural effect. - **Content (contingent):** the *decision* and the *ethical idiom* the framing yields depend on **framing × language × model**. "You decide" → utilitarian/committed; "from all perspectives" → procedural; reflexive → hedged; and Yoruba systematically commits to a sharper moral register than English.

So the headline is not "framing changes the voice" but "**framing changes the verdict and the ethics — and language changes them again.**" A single dilemma can receive a duty-based refusal in impersonal English and a utilitarian commitment in second-person Yoruba *from the same model*.

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## Caveats

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- **Two models only.** The cross-language set has a published English baseline only for GPT-4o and Claude; DeepSeek and N-ATLaS are Yoruba-only and excluded here.
- **Claude version confound.** The Yoruba Claude is **Claude Sonnet 4**, the English Claude is **Claude 3.5 Sonnet** (see methods §A9bis). By-model language divergence (Claude 0.63 vs GPT-4o 0.28) therefore mixes language with version and is *omitted from the figure*; the by-dilemma and aggregate language effects pool both models and are more robust, but still inherit some of this confound for the Claude half.
- **Open-style responses, single coder, small N** (54 cells/model/language). Treat magnitudes as indicative; the directions are the finding.